

ONLY TURN IN SOLUTIONS TO EXERCISES 1 AND 2.

1. Suppose that X is a positive random variable. We know that $E(1/X) \neq 1/E(X)$, in general. Can you claim that either $E(1/X) \geq 1/E(X)$ or $E(1/X) \leq 1/E(X)$? Explain your response.
2. Consider a circle of radius 1 centered at the origin, and suppose that a point within the circle is randomly chosen in such a manner that all regions within the circle of equal area are equally likely to contain the point. (In other words, the point is uniformly distributed within the circle.) If we let X and Y be the coordinates of the point chosen, it follows that the joint density function of X and Y is given by:

$$f_{XY}(x, y) = \begin{cases} 1/\pi & x^2 + y^2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the marginals of X and Y . (They are the same due to symmetry.)
- (b) Let D be the distance between the origin to the point selected. Find the probability that D is less or equal to d . In other words, find $F_D(d)$.

Guidance: Note that $D = \sqrt{X^2 + Y^2}$.

- (c) Find $E(D)$.

3. Consider the following joint pdf:

$$f_{X,Y}(x, y) = \begin{cases} ye^{-y(x+1)} & x > 0, \ y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the density function of $T = XY$, and identify the distribution you got.

Guidance: Find $F_T(t) = P(XY \leq t) = P(Y \leq t/X) = \int \int \dots$ and then differentiate.

Note: One of the two ways to setup the double integration is more “friendly” than the other.

4. Assume that U_1 and U_2 are independent uniform(0, 1) random variables. Let $U_{(1)}$ denote the minimum of U_1 and U_2 , and let $U_{(2)}$ denote the larger of those two. Finally, let

$$L_1 = U_{(1)}, \quad L_2 = U_{(2)} - U_{(1)}, \quad \text{and} \quad L_3 = 1 - U_{(2)}.$$

Note that this is the “stick-broken-into-three-pieces” example, with L_i denoting the lengths of the three pieces ordered from left to right.

In lecture we derived the joint distribution of $(U_{(1)}, U_{(2)})$. You can build on that result here.

- (a) What is the joint distribution (pdf) of (L_1, L_2) ?

Guidance: Use the joint pdf of $(U_{(1)}, U_{(2)})$ as a starting point. Look at the support for $(U_{(1)}, U_{(2)})$ and ask: For what region is $L_1 \leq \ell_1$ and $L_2 \leq \ell_2$?

- (b) Show that L_1 and L_2 both have the Beta(1, 2) distribution. (Do not take it as “obvious” that these have the same distribution. Show the derivation.)

① Let $g(x) = \frac{1}{x}$, A convex function when $x > 0$.
 Then, by Jensen's Inequality, $E(g(x)) \geq g(E(x))$.
 $\therefore \boxed{E\left(\frac{1}{x}\right) \geq \frac{1}{E(x)}}$ when x is positive.

② $f_{X,Y}(x,y) = \begin{cases} \frac{1}{\pi}, & x^2 + y^2 \leq 1 \\ 0, & \text{otherwise} \end{cases}$

$$\begin{aligned} \text{(a) } f_X(x) &= \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} dy \\ &= \frac{1}{\pi} y \Big|_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \\ &= \frac{1}{\pi} \cdot 2\sqrt{1-x^2} \end{aligned}$$

$$= \frac{2}{\pi} \sqrt{1-x^2}, \quad x \in [-1, 1]$$

$$f_X(x) = 0, \quad x \notin [-1, 1]$$

$$\begin{aligned} f_Y(y) &= \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \frac{1}{\pi} dx \\ &= \frac{1}{\pi} x \Big|_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \\ &= \frac{1}{\pi} \cdot 2\sqrt{1-y^2} \end{aligned}$$

$$= \frac{2}{\pi} \sqrt{1-y^2}, \quad y \in [-1, 1]$$

$$f_Y(y) = 0, \quad y \notin [-1, 1]$$

$$\begin{aligned} \text{(b) } F_D(d) &= P(D \leq d) \\ &= P(\sqrt{x^2 + y^2} \leq d) \end{aligned}$$

$$\begin{aligned} &= \iint_{x,y: \sqrt{x^2+y^2} \leq d} \frac{1}{\pi} dx dy \\ &= \frac{1}{\pi} \iint_{x^2+y^2 \leq d^2} dx dy, \quad (\text{Area of a circle with radius } d) \\ &= \frac{1}{\pi} \cdot \pi d^2 \end{aligned}$$

$$= d^2$$

$$\begin{aligned} \text{(c) Cmf: } F_D(d) &= d^2 \\ \therefore \text{pdf: } f_D(d) &= \frac{dF_D(d)}{dd} = 2d \end{aligned}$$

$$\therefore E(D) = \int_0^1 d \cdot 2d \, dd$$

$$\begin{aligned} &= 2 \int_0^1 d^2 \, dd \\ &= 2 \cdot \frac{1}{3} d^3 \Big|_0^1 \end{aligned}$$

$$= \frac{2}{3}$$